NITTANY DATA LABS

Handling Imbalanced Data

Bootcamp #2, Lab 1

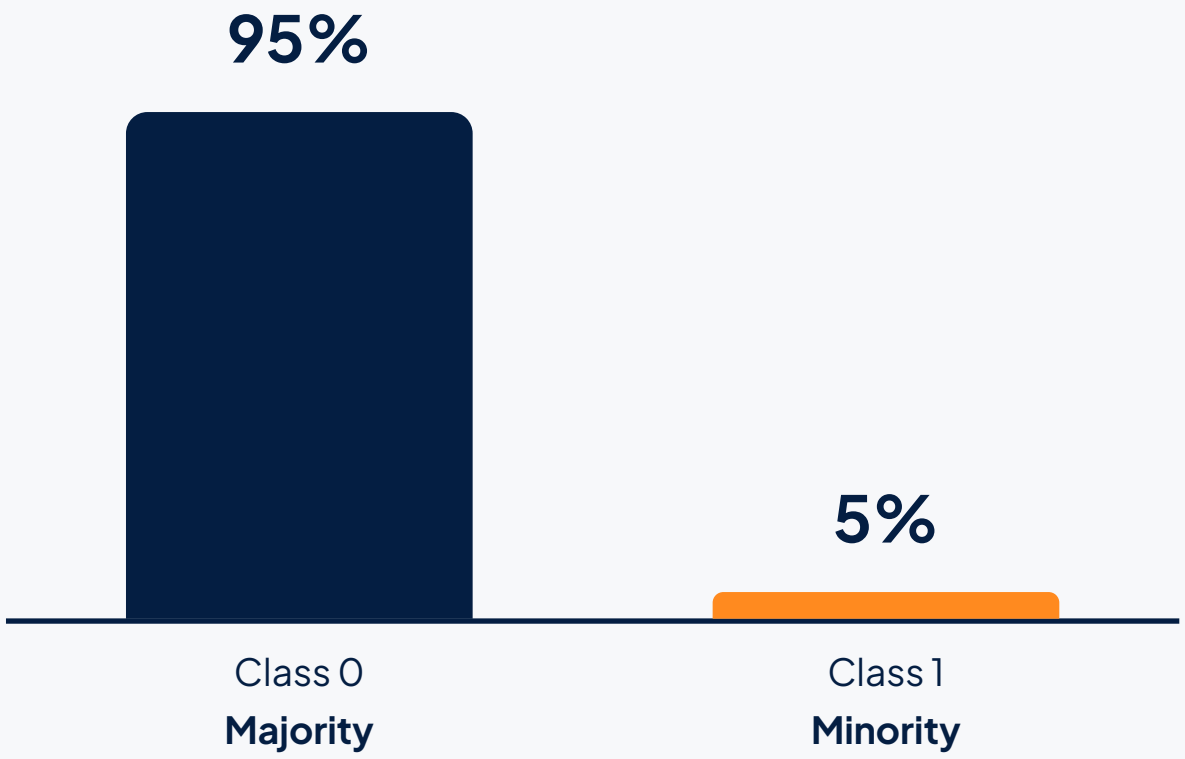


Attendance – “Balance”



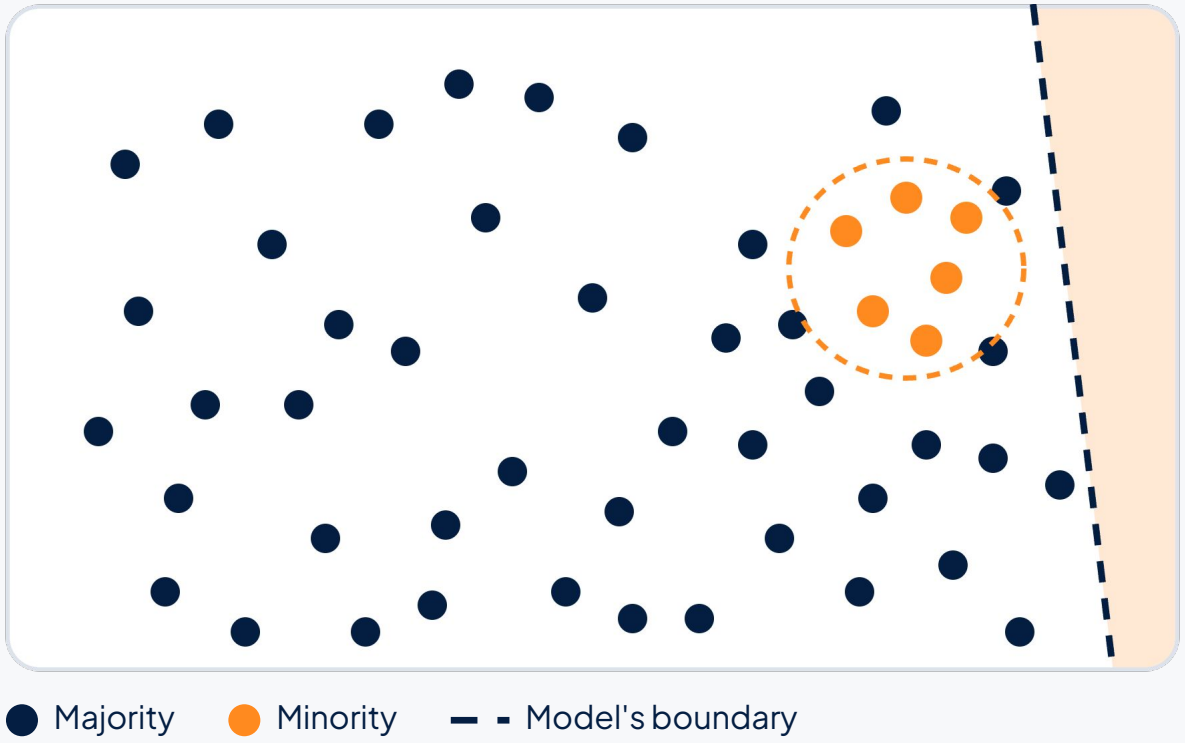
What is imbalanced data?

- One class has **way** more samples
- Typical split: 95% vs 5%
- Fraud, disease diagnosis, spam, equipment failure
- The rare class is usually the one we care about



Why it's a problem

- Models lean toward predicting the majority
- Minority samples get treated like noise
- The decision boundary skews to fit the crowd



The accuracy trap



but

0
fraud caught

A model that always says “not fraud” on data with 1% fraud

Better metrics

Precision

Of everything we flagged, how much was right?

`TP ÷ (TP + FP)`

Recall

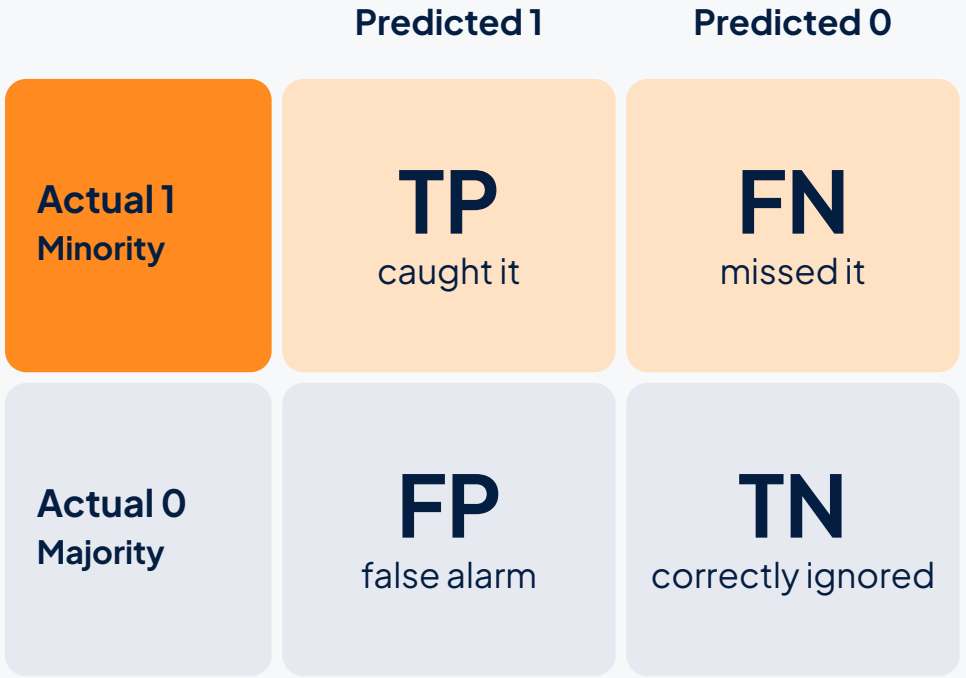
Of all real positives, how many did we catch?

`TP ÷ (TP + FN)`

F1 score

Balances both: only goes up when both go up

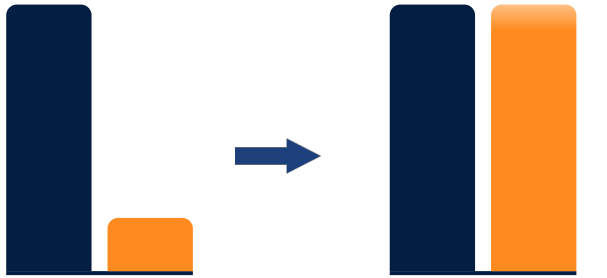
`2 × P × R ÷ (P + R)`



Resampling basics

Random Oversampling

Duplicate minority samples

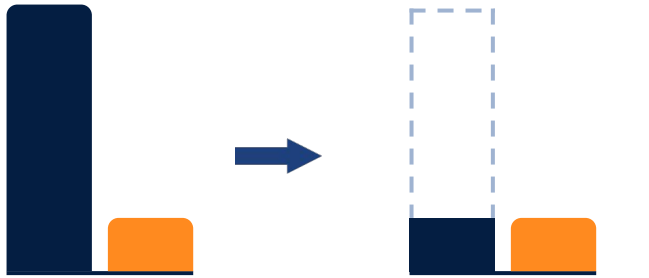


+ Keeps every data point

– Can overfit to the copies

Random Undersampling

Remove majority samples

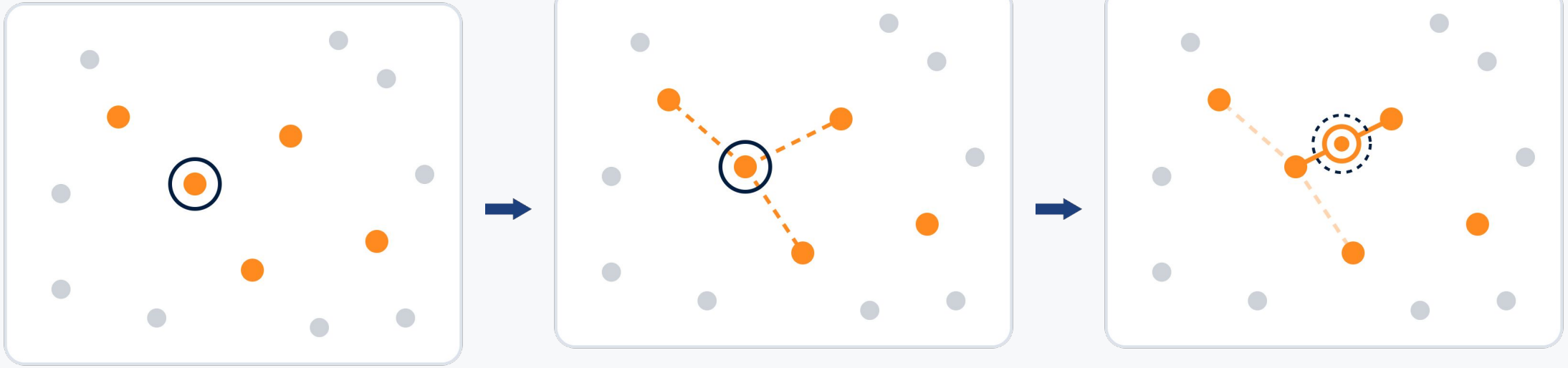


+ Fast and simple

– Throws away useful data

SMOTE: create new points, don't copy

Synthetic Minority Over-sampling Technique



1. Pick a minority point

2. Find its k nearest minority neighbors

3. Place a NEW point somewhere along a line

SMOTE in code

```
from sklearn.model_selection import train_test_split
from imblearn.over_sampling import SMOTE
# 1. Split FIRST
X_train, X_test, y_train, y_test = train_test_split(X, y, stratify=y)
# 2. Resample the TRAINING set only
X_res, y_res = SMOTE(random_state=42).fit_resample(X_train, y_train)
```

⚠ **Only resample the TRAINING set — never the test set.**



SMOTE is not magic

30% → 80%
Recall ↑ fewer missed positives

67% → 42%
Precision ↓ more false alarms

92% → 87%
Accuracy ↓ and that's okay

Before SMOTE

	Pred 1	Pred 0
Actual 1 Minority	6 TP	14 FN
Actual 0 Majority	3 FP	177 TN

After SMOTE

	Pred 1	Pred 0
Actual 1 Minority	16 TP	4 FN
Actual 0 Majority	22 FP	158 TN

Class weights and balanced bagging

- **Class weights:** tell the model that missing an orange point costs more
- **Balanced bagging:** train many small models on balanced slices, then let them vote
- Tree models like Random Forest often handle this better



Sometimes You Shouldn’t Balance

- **Model performance:** if model is already performing well, balancing might be unnecessary
- **Distribution reflecting reality:** if only 1% of transactions are actually fraudulent, balancing changes the original distribution
- **Increases false positives:** if you want to avoid false positives, then balancing might introduce unnecessary weight to minority class

Thank you for coming!

Next Meeting: 11/1 – Evaluation Metrics